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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

MAKE UP EXAMINATIONS – JUNE 2023

Program	:	**B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)**	**Semester**	:	**III**
Course Name	:	**Data Structures**	**Max. Marks**	:	**100**
Course Code	:	**CY33 / CI33**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. a) Give the general syntax to initialize a structure variable during compile time. Define a structure to represent time with following fields hour, min and seconds. Using this structure, develop a C program to read time during compile time and find the total time in seconds. Display the input time and the resultant time in seconds with suitable messages. CO1 (06)

b) Write a program to implement string reverse functionality with a single dimension array. CO1 (06)

c) The department of CSE wants to maintain the details of program toppers of each batch. Define a structure topper to represent the following information: name, USN, CGPA and year of passing. Using this structure, develop a C program to read the details of n toppers and sort them in descending order based on the year of passing. Display the details of n toppers with suitable messages. CO1 (08)

2. a) Write a structure DOB to represent Date of Birth. Write another structure named student with following fields: name and a member of struct DOB type. Using this nested structure, develop a C program to read name and date of birth of two students and check whether both are born on same date or not. Display the result with suitable messages. CO1 (06)

b) Consider a structure Course with following fields: title, semester and number of credits. Develop C functions to perform the following operations on an array of n courses: (i) Display all the courses of a specified semester. (ii) Compute the total credits of all the courses belonging to a specific semester. CO1 (06)

c) Develop a C program with an appropriate structure definition and variable declaration to read and display information about an employee using nested structures. Consider the following fields like Ename, Eid, DoJ(Date, Month, Year) and Salary(Basic, DA, HRA). CO1 (08)

UNIT - II

3. a) Write a C program to implement parenthesis checker using stacks. CO2 (08)

b) Illustrate the process of converting the following infix expressions into postfix expressions using stacks: (A+(B*C)/(D-E)+F-G/H). CO2 (08)

c) Differentiate between iteration and recursion. Write a recursive C program to implement binary search algorithm. CO2 (04)

4. a) Write algorithm to convert an infix expression to a postfix expression using stacks. CO2 (08)

b) Write a C program to implement stack of character's and check if the contents of stack is palindrome or not. CO2 (08)

c) Illustrate the process of converting the following infix expressions into prefix expressions using stacks: ((A-B) * (C+D)/E+F-G). CO2 (04)

UNIT - III

5. a) Explain the process of inserting a new element and deleting an element in a priority queue by taking an example for each. CO3 (10)

b) Write a program in C to implement a linear queue to perform insertion, deletion and display of items in a linear queue. CO3 (10)

6. a) Write a 'C' program to implement multiple queues in the same array of sufficient size. CO3 (10)
(Implement insertion, deletion and display operation in the program)

b) Illustrate the process of inserting an element and deleting an element from a circular queue by taking an example. CO3 (10)

UNIT- IV

7. a) Write C functions for push() and pop() operations for a linked stack. CO4 (06)

b) Write a C program to support following operations on doubly linked list. CO4 (08)
(i) Insert new note to left of node whose key value is read as input
(ii) Delete a node with a given data.

c) Compare singly, circular and doubly linked lists with suitable examples for each. CO4 (06)

8. a) Write a C function to attach one doubly linked list to the end of another. CO4 (08)

b) Write a C program to maintain a stack of integers using linked implementation method. CO4 (08)

c) Write a C program to create a singly linked list that maintains a list of names in alphabetical order. Implement the following operations on the list. CO4 (04)
i) Insert a new name
ii) Delete a specified name.

UNIT - V

9. a) Explain different steps to convert the given general tree as shown in Fig.9(a) to binary tree: CO5 (08)

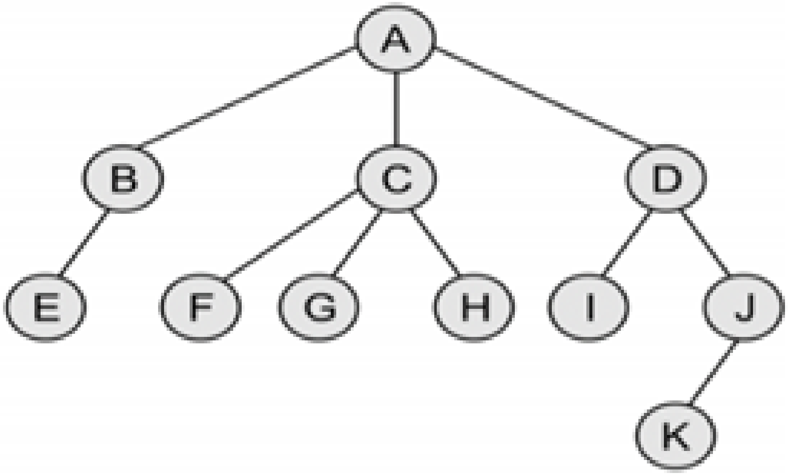

Fig.9(a) Tree

b) Explain how the following binary tree as shown in Fig.9(b) is traversed using inorder, preorder and post order tree traversal technique: CO5 (08)

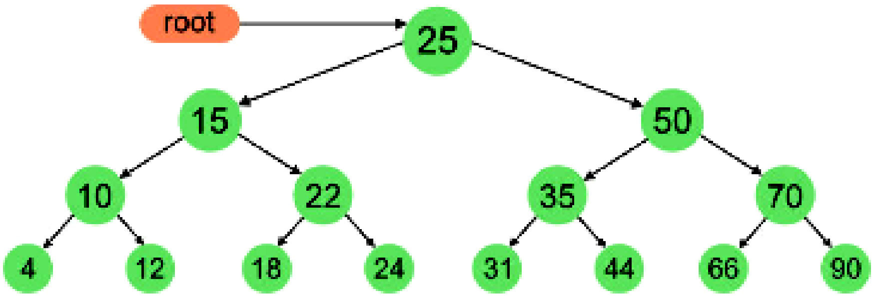

Fig.9(b) Binary tree

c) Evaluate the infix expression represented in the tree as shown in Fig.9(c), given a = 30, b = 10, c = 2, d = 30, e = 10. CO5 (04)

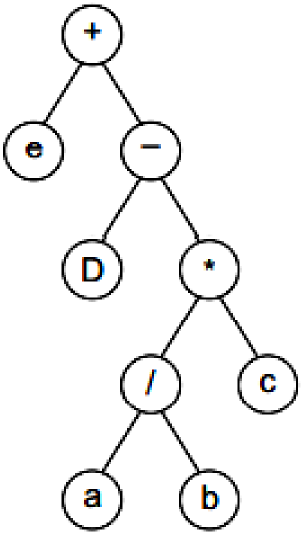

Fig.9(c)

10. a) Consider the tree given in Fig.10(a). Find the in-order, pre-order, post-order, and level order traversal. CO5 (08)

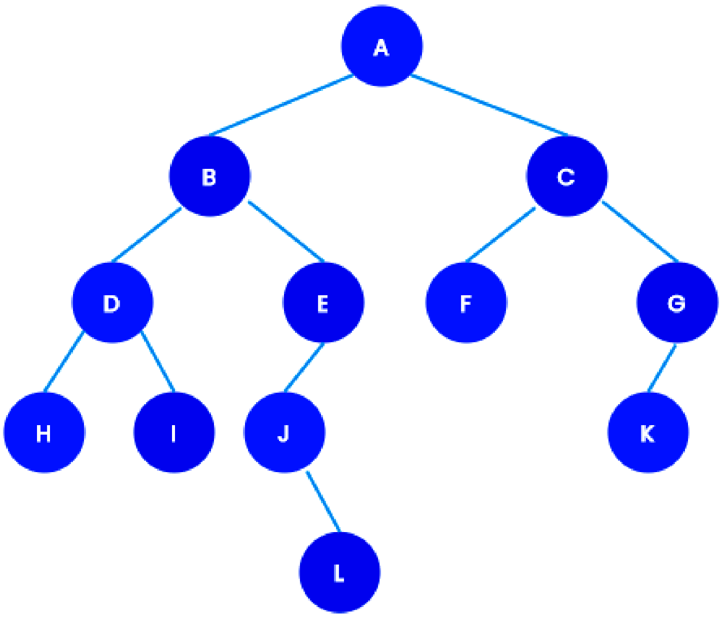

Fig.10(a) Tree

b) Given an expression, Exp = ((a + b) – (c * d)) % ((e ^f) / (g – h)), construct the corresponding binary tree. CO5 (08)

c) Draw the memory representation of the binary tree given in Fig.10(c). CO5 (04)

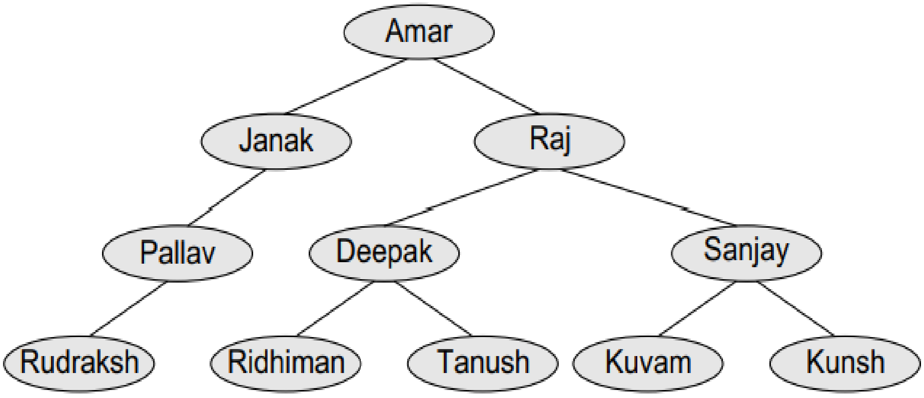

Fig.10(c) Binary tree

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